

Indian Institute of Information Technology Bhopal

Department of Computer Science & Engineering

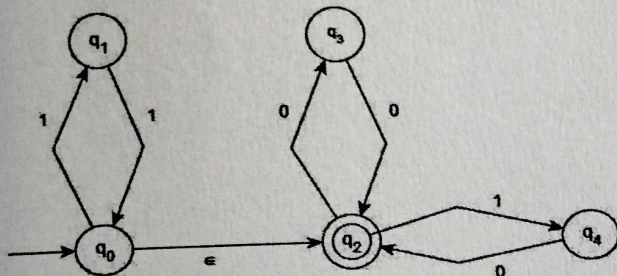
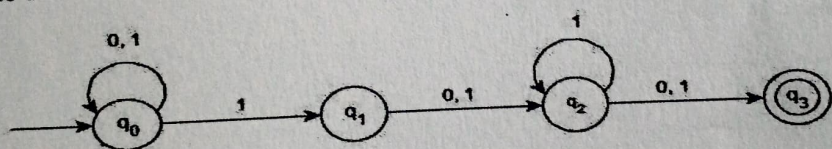
End Term Examination – December 2023

Subject Code:- CSE-225**Subject Name:- Formal Language and Automata Theory****Time:- 03:00 Hours****Maximum Marks: 60****Note:**

1. All the questions are compulsory and while solving write question number and its part clearly.

2. All questions carry equal marks.

3. Solve questions in given sequence and all the sub parts should be at same place.

Q. No.	Questions	Marks
1	(a) Explain the difference between a deterministic finite automaton (DFA) and a nondeterministic finite automaton (NFA). Provide an example for each. (b) Define the pumping lemma for regular languages. How is it used to show that a language is not regular? Provide an example.	12
2	(a) Convert the ϵ -NFA into minimal DFA  (b) Consider the finite automaton in the following figure.  What is the set of reachable states for the input string 0011? With Explain (A) $\{q_0, q_1, q_2\}$ (B) $\{q_0, q_1\}$ (C) $\{q_0, q_1, q_2, q_3\}$ (D) $\{q_3\}$ (c) How formal language is different from natural language.	12
3.	(a) Construct minimal DFA with 5 tuples that accepts all a's and b's where both no. of a's and b's are divisible by 2. (b) Explain the concept of decidability in the context of formal languages and automata. (c) Consider the following grammar: $S \rightarrow AB$ $A \rightarrow a$ $B \rightarrow C/b$ $C \rightarrow D$ $D \rightarrow a$ $E \rightarrow a$	12

	Remove the unit productions from the said grammar.	
4.	(a) Design a context-free grammar for a simple arithmetic expression language that includes addition, subtraction, multiplication, and division. (b) Given a regular expression $(0 + 1)^*$, design a deterministic finite automaton (DFA) that recognizes the language described by the regular expression. (c) Design a PDA for the following language L. $L = \{a^n b^m c^{n+m} \mid n, m \geq 1\}$	12
5.	(a) Discuss the importance of the Chomsky hierarchy in classifying formal languages. Provide examples for each language class and explain their computational power. (b) Calculate leftmost derivative and rightmost derivative? Prove that the following grammar is ambiguous. $E \rightarrow E + E / E \times E / 2 / 3 / 4$ (c) Design a Turing machine as a comparator, where $\Sigma = \{1\}$, $\Gamma = \{1, 0, X, B\}$	12